

Selenium Actions Class - Interview Revision Notes

What is Actions Class?

Actions class is used to perform advanced mouse and keyboard operations in Selenium.

Import Statement

```
import org.openqa.selenium.interactions.Actions;
```

Object Creation

```
Actions actions = new Actions(driver);
```

1. Mouse Hover

Used to move the mouse pointer over an element.

```
Actions actions = new Actions(driver);  
actions.moveToElement(element).perform();
```

2. Right Click (Context Click)

Used to perform a right-click operation.

```
Actions actions = new Actions(driver);  
actions.contextClick(element).perform();
```

3. Double Click

Used to double-click on an element.

```
Actions actions = new Actions(driver);
actions.doubleClick(element).perform();
```

4. Click

Used when normal WebElement click() is not working.

```
Actions actions = new Actions(driver);
actions.click(element).perform();
```

5. Click and Hold

Used to click and hold the mouse button on an element.

```
Actions actions = new Actions(driver);
actions.clickAndHold(element).perform();
```

6. Release

Used to release the mouse button after clickAndHold().

```
Actions actions = new Actions(driver);
actions.release().perform();
```

7. Drag and Drop

Used to drag an element and drop it onto another element.

```
Actions actions = new Actions(driver);
actions.dragAndDrop(source, target).perform();
```

8. Drag and Drop by Offset

Used to drag an element by specified x and y coordinates.

```
Actions actions = new Actions(driver);
actions.dragAndDropBy(element, 100, 50).perform();
```

Click and Hold + Move + Release

```
Actions actions = new Actions(driver);

actions.clickAndHold(source)
    .moveToElement(target)
    .release()
    .perform();
```

Build and Perform

```
Actions actions = new Actions(driver);

actions.moveToElement(element)
    .build()
    .perform();
```

Note:

- perform() executes the action.
- build() creates a composite action before execution.
- In Selenium 4, build() is usually optional.

Frequently Asked Interview Methods

1. moveToElement() -> Mouse Hover
2. contextClick() -> Right Click
3. doubleClick() -> Double Click
4. click() -> Click

- 5. clickAndHold() -> Hold Mouse Button
 - 6. release() -> Release Mouse Button
 - 7. dragAndDrop() -> Drag and Drop
 - 8. dragAndDropBy() -> Drag by Coordinates
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Interview Answer

Actions class is used to perform advanced user interactions such as mouse hover, right-click, double-click, drag-and-drop, click-and-hold, and keyboard actions. It is part of Selenium WebDriver and helps automate complex user gestures.